

Name | Roll number

Indian Institute of Technology Madras
Placement Registration number :-



Education

<u>Program</u>	<u>Place of study</u>	<u>Years of study</u>	<u>Percentage scored</u>
B Tech, Electrical Engineering – proof	Indian Institute of Technology, Madras	2017-2021	CGPA :- x.xx
XII – proof	Sri Chaitanya Junior College, Hyderabad	2016-2017	xx.xx% (State Syllabus)
XI – proof	Sri Chaitanya Junior College, Hyderabad	2015-2016	xx.xx% (State Syllabus)
X – proof	Narayana Techno School, Hyderabad	2014-2015	GPA :- x.x (State Syllabus)

Scholastic Achievements

- Secured All India Rank **xxx** in **IITJEE 2017** | Percentile-**top xxx** | Scored **xxx** out of **xxx** | proof
- Secured All India Rank **xxx** in **JEE Mains Paper-1 2017** | Percentile-**top xxx** | Scored **xxx** out of **xxx** | proof
- Secured All India Rank **xxx** in **JEE Mains Paper-2 2017** | Scored **xxx** out of **xxx** | proof
- Secured State Rank **xxx** in **TS-EAMCET 2017** (State exam) | Percentile-**top xxx** | Scored **xxx** out of **xxx** | proof

Professional Experience

Software Development Intern – xxxxxxxxxxxxxxxxx - certificate

HYDERABAD, INDIA MAY-JUL 2020

Mentor: xxxxxxxx

- Worked on Universal Verification Methodology (UVM) Framework and developed a python code, aimed to speed up the triage work and get the root cause done by all team members quickly.
- Team members will be sending different types of packets to check the performance of the chip.
- Different types of packets are sent from different interfaces and the goal is to trace the life cycle of the packet and report errors packets which lack in information.
- The key challenge was to ensure that all error packets are detected for any given interface.

Course Work

- Electrical** – Digital systems, Signals and Systems, Microprocessors, Computer Organisation, Electromagnetics, Analog Systems, Analog Circuits, Digital Signal processing, Solid State Devices, Electric Machines, Digital System Testing & Testable Design, Communication Systems, Digital IC design*.
- Coding** – C programming, Applied Programming, Data Structures and Algorithms, Object Oriented Programming*, Pattern Recognition and Machine Learning, Fundamentals of deep learning.
- Maths** – Graph theory, Probability, Series and Matrices, Functions in Several Variables.

Course and Lab projects

CS5691: Pattern Recognition and Machine Learning: Prof. L A Prashanth

IIT MADRAS JAN-MAY 2020

- Coded Machine learning algorithms like Bayes Classifier, Maximum Likelihood Estimation, SVD, Linear Regression, Kernel Regression, Logistic Regression, Perceptron Classifier many more.
- Coded PCA algorithm and used it for Image Reconstruction and used SVM's algorithms to classify data.
- Used **Python** language to code algorithms mentioned above from scratch.

EE4371: Data Structures and Algorithms: Prof. Harishankar Ramachandran**IIT MADRAS JUL-NOV 2019**

- Coded many assignments which included solving questions like Traffic intersection using greedy algorithm, Stack-based postfix calculator, different kinds of Knapsack problem using dynamic programming and also implemented event simulation and devices like router using priority queues.
- Used **C** and **C++** languages to code algorithms mentioned above from scratch.

EE2003: Computer Organisation: Prof. Nitin Chandrathoodan**IIT MADRAS JUL-NOV 2019**

- Build from scratch, a CPU which uses the Risk-V instruction set architecture.
- Designed and Implemented a Pipelined, Error-handling and Peripheral-handling CPU in **Verilog** and used it as CPU on Xilinx spartan 3E board and tested it.

EE3002: Analog Circuits: Prof. S Aniruddhan**IIT MADRAS JUL-NOV 2019**

- Designed and Implemented a Two-Stage nMOS Opamp with required specifications under given conditions.
- Used **LTspice** for simulation mentioned above from scratch.

EE2019: Analog Systems: Prof. S Aniruddhan**IIT MADRAS JAN-MAY 2019**

- A member in a duo to design very effective band pass filter which gives out a light and a speaker output when a certain frequency is given as input.
- Used **LTspice** for simulation mentioned above from scratch.

EE2703: Applied Programming Lab: Prof. Harishankar Ramachandran**IIT MADRAS JAN-MAY 2019**

- Coded many assignments which include solving many complex circuits, dealing with finding the responses given by analog circuits and plotting them and many more.
- Used **Python** language to code algorithms mentioned above.

EE2016: Microprocessor Theory: Prof. R Manivasakan**IIT MADRAS JUL-NOV 2018**

- Coded many assignments in **Assembly** which included prime factorisation, taking input message from the board and also controlling devices, basic array operations, multiplications and divisions and many more.

EE2001: Digital Systems: Prof. Vinita Vasudevan**IIT MADRAS JAN-MAY 2018**

- A member in a duo to operate FGPA by coding in Verilog and also designed many combinational and sequential circuits
- Used **Verilog** language to code algorithms mentioned above.

Skills

- **Programming languages** – C , Python , C++ , Java , Verilog , System Verilog , Assembly coding for Micro processors.
- **Framework** – Universal Verification Methodology (UVM) Framework.
- **Tools** – LaTeX , MATLAB , Octave , Eagle , LTspice , Xilinx ISE , Jupyter.

Position of Responsibility

Tech-Creation Coordinator, Team Shaastra**IIT MADRAS SEPT 2018 – JAN 2019**

- Part of **Shows and Exhibitions** as a co-ordinator. Worked in a team and created and built 10 different games using scrap material.
- Managed a huge crowd of students who where enthusiastic to play these games.

Extra-Curricular Activities

- Among 25 students selected from batch 2017 for **IITM Ultimate** under **National Sports Organisation**.

Skills and Hobbies

- Playing Volleyball, Cricket, Badminton, etc.
- **PC Gamer**
- Can solve different types of Rubik's cubes.
- **Cyclist**